

Space Missions and Systems

Orbit Determination Challenge – AA 2019-2020

A spacecraft is in an equatorial orbit around Jupiter. Two flybys of the Galilean moons Ganymede (flyby 1) and Europa (flyby 2) have been planned by the flight dynamics team. Less than one day before the first moon encounter a major failure in the spacecraft tracking equipment made range and range-rate TX/RX capabilities unavailable.

The last estimate, based on range and range-rate data, provided by the navigation team is:

State estimate based on last navigation data available at $t = 0$		
Parameter	Value	Uncertainty
X	1180486.0 km	1 km
Y	0.6 km	1 km
V_x	0.0 km/s	1 m/s
V_y	8.330 km/s	1 m/s
$GM_{Ganymede}$	9887 km ³ /s ²	1 km ³ /s ²

The estimate is provided in a Jupiter-centered inertial reference frame.

The only available measurements are those provided by the on-board camera. The camera is able to provide the relative direction of the moons with respect to the spacecraft position. All the measurements are referenced to the Jupiter-centered inertial reference frame.

A schematic representation of the problem geometry is presented in figure 1.

Immediately after the Ganymede flyby a refined orbit reconstruction is required to assess the impact probability with Europa. The criterion set for deciding whether to perform an orbit correction maneuver is the following

No Maneuver Needed	Maneuver Needed
$7000 \text{ km} < CA < 8000 \text{ km}$	$CA < 7000 \text{ km}$ or $CA > 8000 \text{ km}$

Where CA is the closest approach radial distance to Europa.

- 1) With the provided Ganymede optical observations determine the minimum variance estimate of the state of the spacecraft and the GM of Ganymede at the reference time $t = 0$
- 2) Assess the uncertainty on the closest approach radial distance to Europa and determine if an orbit correction maneuver is needed
- 3) Compute and plot the uncertainty ellipse of the position of the spacecraft at the closest approach with Ganymede before and after processing the optical data.

After the Ganymede flyby, Europa optical observations are collected to further refine the orbit reconstruction:

- 4) Determine the minimum variance estimate of the state of the spacecraft and the GM of Ganymede at the reference time $t = 0$ with the full dataset. Comment the results.

- 5) Comment the new estimate of the closest approach radial distance of the spacecraft to Europa obtained with both datasets.
- 6) Planetary protection rules require $< 0.5 \%$ impact probability in a square region of side $(2R_{moon} + 400\text{km})$ centered on the target. Comment the variation of the impact probability prior to the optical observation campaign and after.

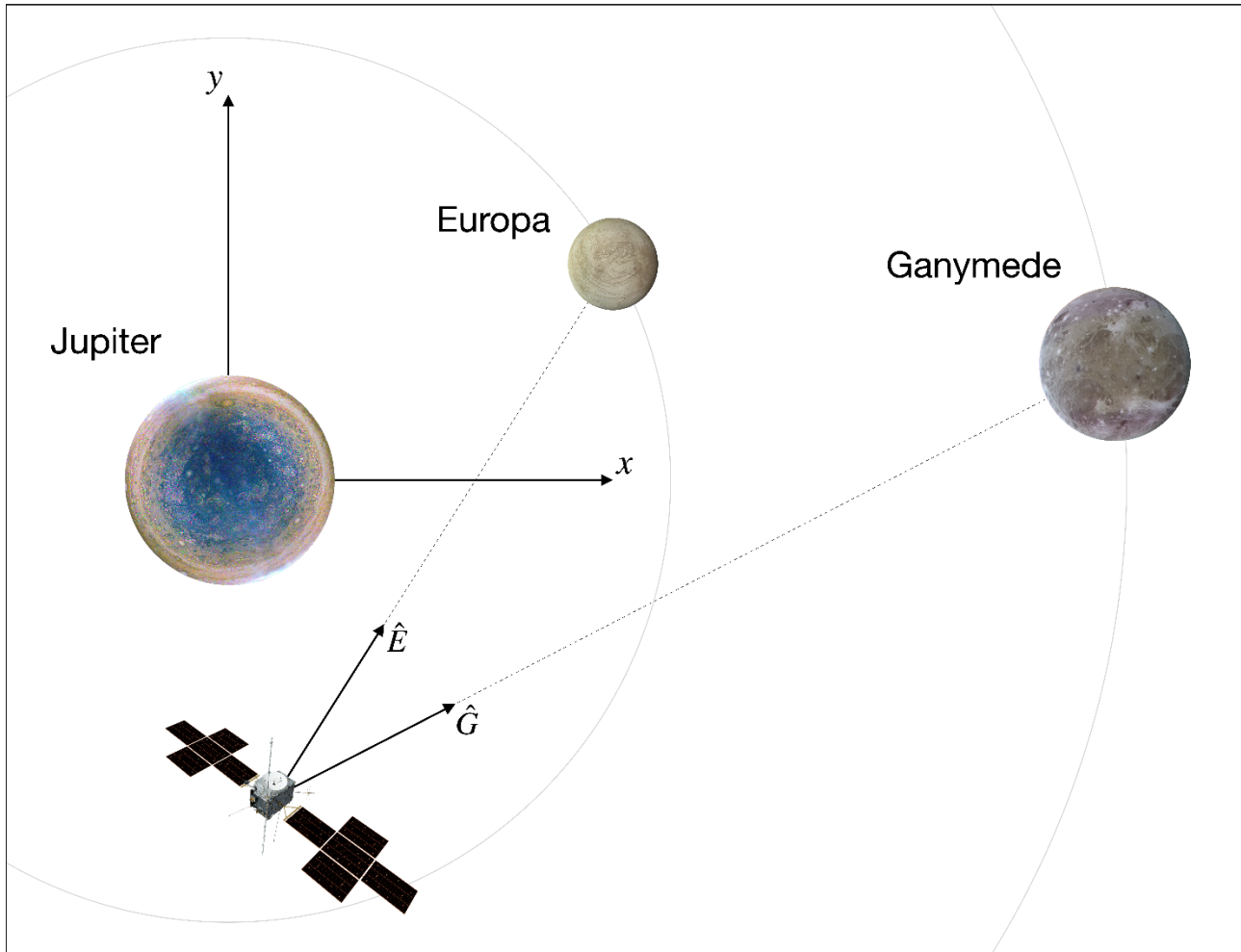


Figure 1 – Schematic representation of the problem geometry. The inertial reference frame (x, y) is Jupiter-centric. The on-board optical camera provides the spacecraft-moon unit vectors $(\hat{E}, \hat{G})$ for Europa and Ganymede respectively)

Additional information:

- Consider the orbits of Europa and Ganymede as co-planar with Jupiter's equatorial plane. The orbits are assumed to be circular and perfectly known. At time $t = 0$ the true anomalies of the moons are the following:

$$\begin{aligned} \nu_{Europa} &= -44^\circ \\ \nu_{Ganymede} &= -13^\circ \end{aligned}$$

and the semi-major axes:

$$\begin{aligned} a_{Europa} &= 670900 \text{ km} \\ a_{Ganymede} &= 1070400 \text{ km} \end{aligned}$$

- The spacecraft is subject to the gravitational force of Jupiter, Europa and Ganymede. The gravitational constants (perfectly known) of these bodies are:

$$GM_{Jupiter} = 126686534.20 \text{ km}^3/\text{s}^2$$

$$GM_{Europa} = 3202.74 \text{ km}^3/\text{s}^2$$

- The optical observations are provided in the files optical_ganymede.txt and optical_europa.txt in the form of the unit vectors from the spacecraft to the target body. Each row contains: epoch (first column), x component of the unit vector (second column), y component of the unit vector (third column)
- For integration purposes on MATLAB use ode113 with tolerances RelTol = 1e-13 and AbsTol = 1e-13

Upload a working computer code (Matlab or any other programming language) and a short, but sufficiently explanatory, note (a **pdf** file) with the mathematical procedure, results, comments, and figures on the Google Classroom of the course.

Should you have problems in uploading the files on Google Classroom send the files by email to luciano.iess@uniroma1.it **AND** gael.cascioli@uniroma1.it

The deadline is **26 April 23:59 CEST**.

Be concise and go straight to the point.

In the note, indicate your **first name**, **last name**, and **student id** (a.k.a., “numero di matricola”).

Note: Students who have failed HW1 need to answer correctly to question 1).